STATS 401 · FINAL PROJECT · CS201 HOMEWORK PROCESS

The score *has a past*

Homework scores cluster near full marks. We followed **3,359** online-judge submissions to see how differently students get there, and whether the path travels with the exam.

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Zhengxiang Liu visualization design, D3 implementation

Watch it move

The playable data film, all twelve figures, tables and methods

THE QUESTION

The gradebook shows a 100. In **36** of **43** problem slots, every student who attempted the problem reached full marks in the end. What the 100 does not show is the first try, the retries, or the hour they arrived.

3,359 submissions inside five homework windows

39 students, 43 problem slots

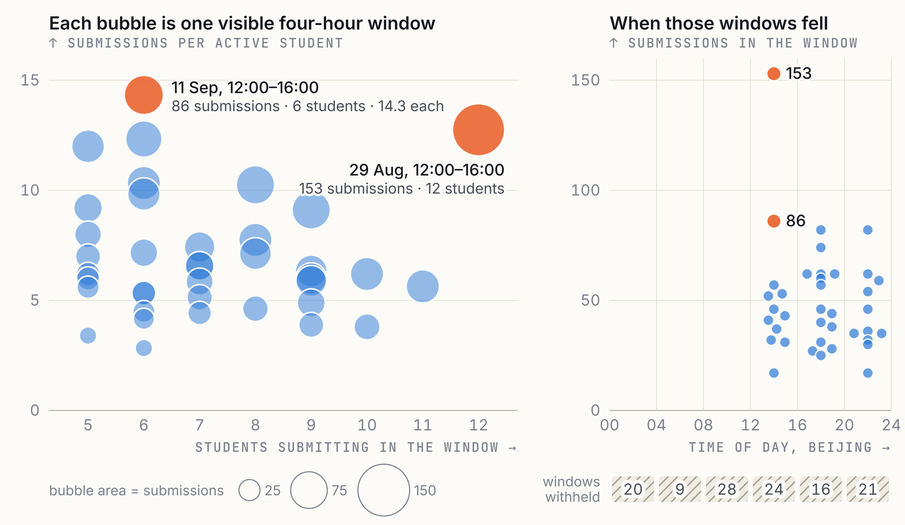
1,345 eligible retries; 547 set a new best

26 midterm grades, a separate pre-exam sample

1 WHEN THE WORK ARRIVES

Film · Figure 01

A burst can be many students, or a few students many times.



Each bubble is one four-hour window in which at least five students submitted. Further right, more students were active; higher up, each of them submitted more often.

The largest visible window holds **153** submissions from **12** students. The most intense holds **86** from only **6**: **14.3** each.

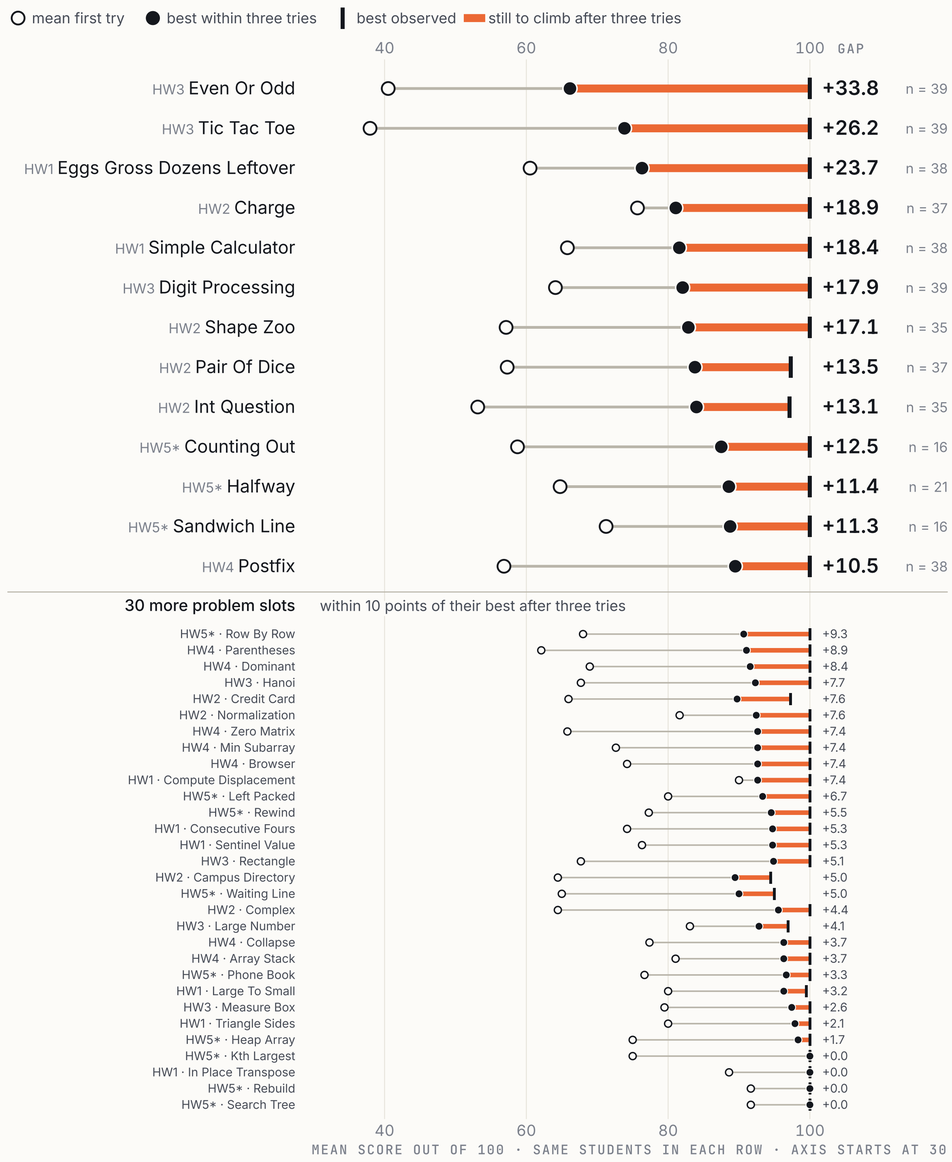
> **1,485** submissions sit in 118 windows with fewer than five students. They are withheld for privacy: counted, never placed, never zero. Clock time is not study time.

2 WHAT THE 100 HIDES

Film · Figure 03

After three tries, 13 of 43 problems still trail their best score by more than 10 points.

For every problem, the same students' mean first try, their best within three tries, and their best observed score. The orange stretch is the distance still to climb after three tries: up to **33.8** points for Even Or Odd.

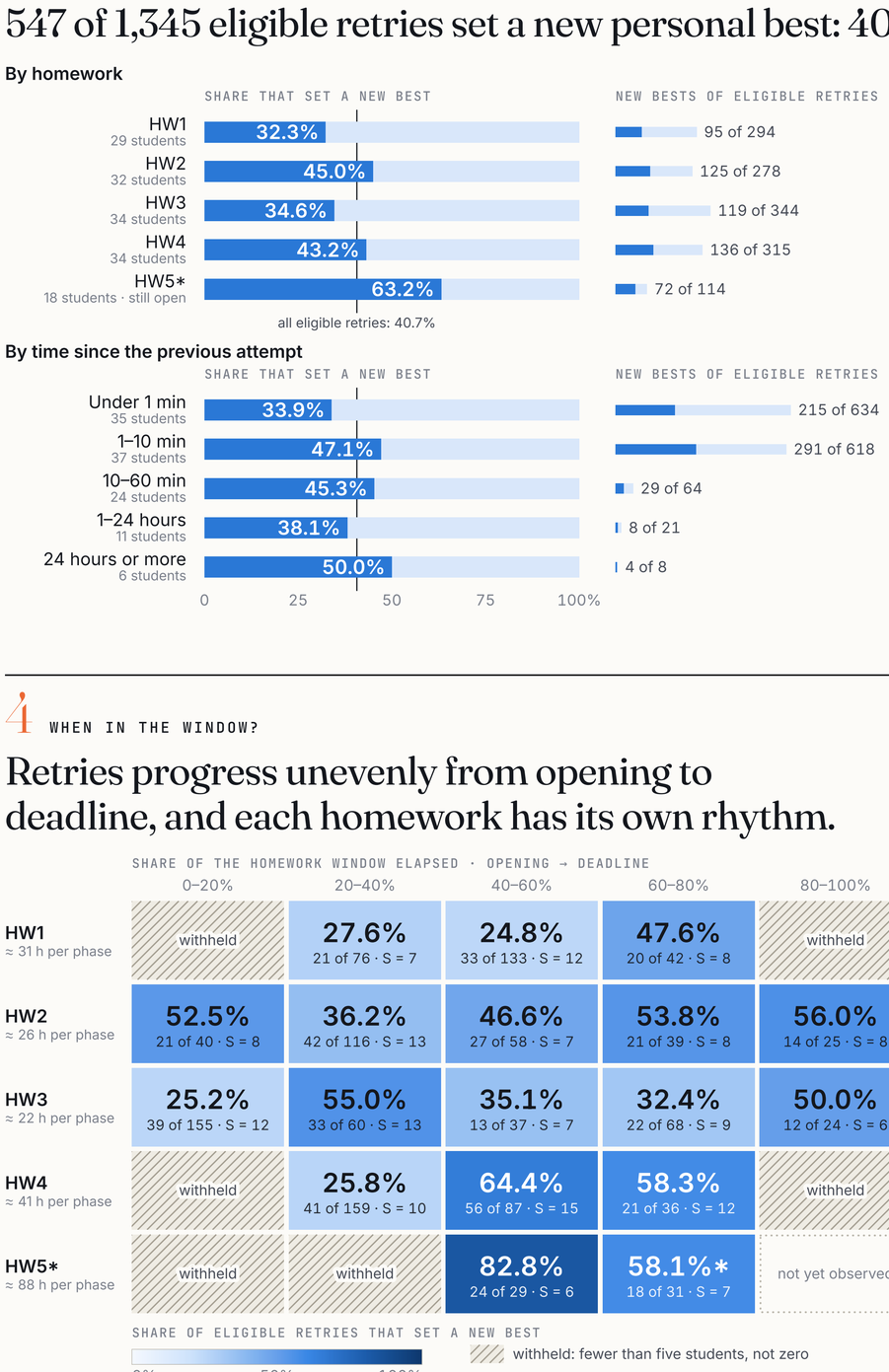


> "Best within three" is a cumulative best. It is not the score of the third submission, and not a causal gain.
> * HW5 is still open: its rows are the scores observed so far.

3 DO RETRIES PAY OFF?

Film · Figure 11

547 of 1,345 eligible retries set a new personal best: 40.7%.



An eligible retry is a scored resubmission made while the student's best on that problem is still below 100. A new best must strictly beat every earlier score.

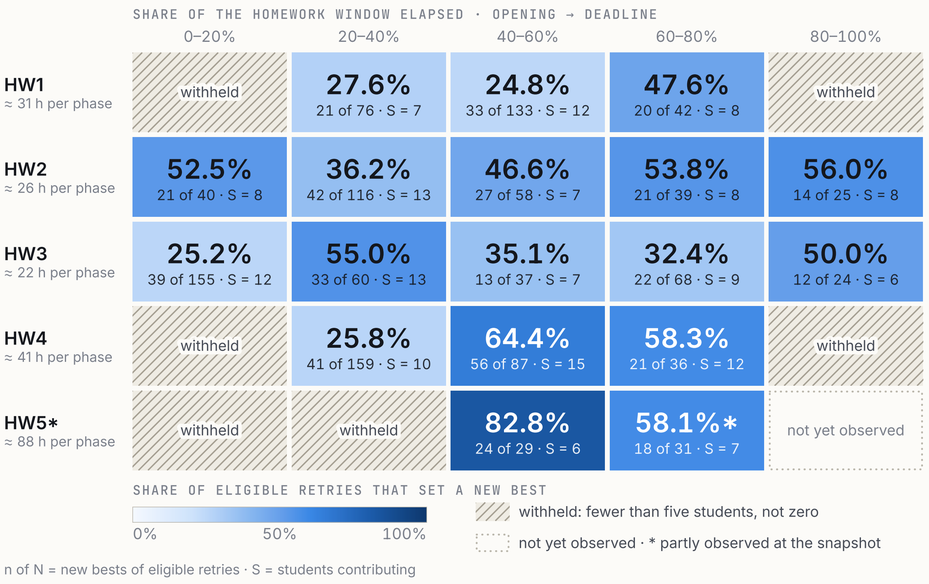
47% of these retries came less than a minute after the previous attempt, and they improved least often: **33.9**%, against **47.1**% after 1–10 minutes.

> Retries are repeated events from the same students, not independent trials. Gaps are clock time, not study time, so these rates do not estimate an effect of waiting. Long gaps have few retries: read the counts.

4 WHEN IN THE WINDOW?

Film · Figure 12

Retries progress unevenly from opening to deadline, and each homework has its own rhythm.



n of N = new bests of eligible retries · S = students contributing

Each homework window is cut into five equal phases. A cell shows the share of that phase's eligible retries that set a new best, with its counts and *S*, the students behind them.

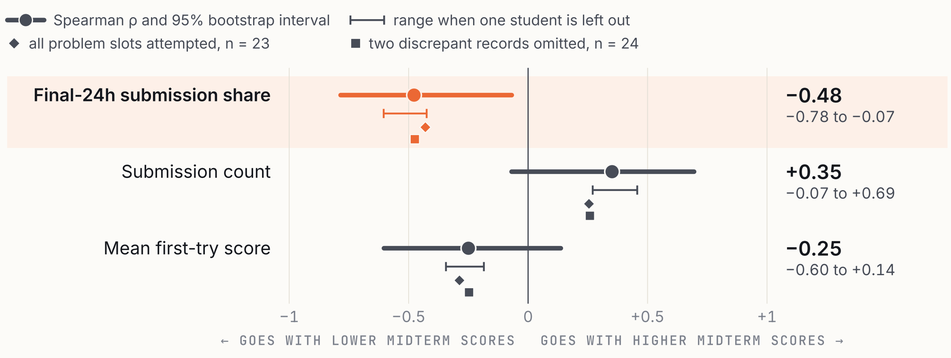
Visible rates run from **24.8**% to **82.8**%. No single pattern repeats across the homeworks.

> Hatched cells have fewer than five students: withheld, not zero. A phase lasts 22–88 hours depending on the homework. Deadlines are the configured ones; individual extensions are unknown. No deadline effect is estimated.

5 DOES IT TRAVEL WITH THE EXAM?

Film · Figure 06

Students who sent less of their work in the final 24 hours tended to score higher on the midterm.



Among **26** students with a supplied grade, the share of homework submissions sent in the last 24 hours before a deadline has Spearman ρ = **−0.48** with the midterm score (95% bootstrap interval −0.78 to −0.07). It is the only interval that excludes zero, and it stays between −0.60 and −0.42 when any one student is left out.

> Grades were supplied for 26 of 41 eligible students: a partial, non-random sample, using pre-exam HW1–HW3 only. An association in a small group, not a cause and not a prediction.

What the *100* was hiding

One score, many roads.

The same full mark is reached in very different ways: windows crowded with students and windows where a few submit again and again; problems nearly solved on the first try and problems still short after three.

Retries progress unevenly.

40.7% of eligible retries set a new best. The rate differs by homework, by how soon a retry follows the last attempt, and by where in the window it falls.

Timing travels with the exam.

In 26 supplied grades, a lower final-day share went with higher midterm scores (ρ −0.48). An association, not a cause.

READ WITH CARE

- Observational data: associations, not causes or predictions.
- Submissions are not effort. Time between submissions is not study time.
- Windows and groups with fewer than five students are withheld, never shown as zero. Only aggregates leave the private data.
- HW5 is still open until 18 Oct: its numbers are observations so far.
- The exam sample is 26 of 41 students, with pre-exam homework only.

DATA Read-only snapshot of the CS201 online judge, 13 Oct 2026, 14:13 Beijing time. HW1–HW5 configured homework windows. Public aggregates only: no names, IDs or individual scores. Midterm grades were supplied separately and linked privately. Charts are drawn with D3 from the same aggregates as the data film.

STATS 401 Final Project
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cis-jujube.github.io/STATS-401-Final-Project